If you want the index of the last element of the string that matches, use the method rindex. The first argument to rindex behaves exactly the same except the return value is based on the last match. In the case of a regular expression, last match means "starts as late in the string as possible" not "ends as late in the string as possible." For example:

```
"The pickaxe book".index("e")    # => 2
"The pickaxe book".rindex("e")   # => 10
"The pickaxe book".rindex(/o.*/) # => 14
```

The index arguments for rindex behave a little differently. They delimit the ending point for the search, which is to say they indicate the maximum index the result can have before returning nil. A positive argument indicates the maximum index directly, while a negative argument implies the maximum index based on adding the negative offset to the length of the string:

```
"The pickaxe book".rindex("e")     # => 10
"The pickaxe book".rindex("e", 2)  # => 2
"The pickaxe book".rindex("e", 4)  # => 2
"The pickaxe book".rindex("e", 8)  # => 2
"The pickaxe book".rindex("e", -7) # => 2
```

If all you want is a boolean yes/no about the substring, you can use include?, which only takes a string argument and returns true if the argument is in the string and false otherwise. For regular expressions, you use match?:

```
"The pickaxe book".include?("e")  # => true
"The pickaxe book".include?("z")  # => false
"The pickaxe book".match?(/o{2}/) # => true
```

If you specifically want to test one end or another of a string, you can use start_with? or end_with?. Somewhat weirdly, these methods take different sets of arguments. The start_with? method takes one or more strings or regular expressions and returns true if the start end of the string matches any of the arguments. The end_with? is the same but does not take regular expression arguments:

```
"The pickaxe book".start_with?("e")     # => false
"The pickaxe book".start_with?("T")     # => true
"The pickaxe book".start_with?(/[A-Z]/) # => true
"The pickaxe book".start_with?("e")     # => false
"The pickaxe book".end_with?("k")       # => true
"The pickaxe book".end_with?("k", "q")  # => true
```

Retrieving Substrings

In Ruby's library, a substring of a string is called a *slice*. The most common way to retrieve a slice from a string is to use square brackets. Five kinds of arguments may be used inside the square brackets:

- A single integer index, as in "abcdefg"[3] or "abcdefg"[-3]. An index of zero will return the first character of the string and a positive number will return the string at that index from the left, so "abcdefg"[3] is d. A negative number will return the index from the right, with -1 being the last character of the string, so "abcdefg"[-3] is e.
- A range, as in "abcdefg"[1..3]. This returns the substring that starts at the index of the beginning of the range and ends at the index of the end of the range. If both parts of the